

The Importance of Feminist Critique for Contemporary Cell Biology

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Biology is seen not merely as a privileged oppressor of women but as a co-victim of masculinist social assumptions. We see feminist critique as one of the normative controls that any scientist must perform whenever analyzing data, and we seek to demonstrate what has happened when this control has not been utilized. Narratives of fertilization and sex determination traditionally have been modeled on the cultural patterns of male/female interaction, leading to gender associations being placed on cells and their components. We also find that when gender biases are controlled, new perceptions of these intracellular and extracellular relationships emerge.

NANCY TUANA HAS TRACED the seed-and-soil analogy from cosmological myths through Aristotle into the biology of the 1700s. Modeling his embryology after his social ideal, Aristotle promulgated the notions of male activity versus female passivity, the female as incomplete male, and the male as the real parent of the offspring. "The female merely provided passive matter to be molded by the male sperm. While there were competing views of embryology during Aristotle's time, Aristotle's principles got the support of St. Thomas and were given the sanction of both religion and scientific

philosophy (Horowitz 1976, 183). In this essay, we will attempt to show that this myth is still found in the core of modern biology and that various "revisionist" theories have been proposed within the past five years to offset this myth.

We have come to look at feminist critique as we would any other experimental control. Whenever one performs an experiment, one sets up all the controls one can think of in order to make as certain as possible that the result obtained does not come from any other source. One asks oneself what assumptions one is making. Have I assumed the

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Lest anyone believe that this is strictly an academic exercise, the *New York Times* (25 March 1987, Sec. 1, p. 20) recently reported an article where in Adrianus Cardinal Simonis, Primate of the Netherlands, cited fertilization as evidence for the passive duties of women. In this essay, the Archbishop pointed to the egg that merely "waits" for the male's sperm, which he described as the "dynamic, active, masculine vector of new life."

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temperature to be constant? Have I assumed that the pH doesn't change over the time of the reaction? Feminist critique asks if there may be some assumptions that we haven't checked concerning gender bias. In this way feminist critique should be part of normative science. Like any control, it seeks to provide critical rigor, and to ignore this critique is to ignore a possible source of error.

The following essay is not an attempt to redress past injustices which biology has inflicted upon women. This task has been done by several excellent volumes that have recently been published (Sayers 1982; Bleier 1984, 1986; Fausto-Sterling 1985). Rather, this paper focuses on what feminist critique can do to strengthen biology. What emerges is that gender biases do inform several areas of modern biology and that these biases have been detrimental to the discipline. In other words, whereas most feminist studies of biology portray it—with some justice—as a privileged oppressor, biology has also been a victim of the cultural norms. These masculinist assumptions have impoverished biology by causing us to focus on certain problems to the exclusion of others, and they have led us to make particular interpretations when equally valid alternatives were available.

SPERM GOES A' COURTIN'

If Aristotle modeled fertilization and sex determination on the social principles of his time, he had plenty of company among more contemporary biologists. The first major physiological model of sex determination was proposed in 1890 when Sir Patrick Geddes and J. Arthur Thomson published *The Evolution of Sex*, one of the first popular treatises on sexual physiology. By then, it had been established that fertilization was the result of the union of sperm and egg. But still unanswered was the mechanism by which this event constructed the embryo. One of the central problems addressed by this highly praised volume was how sex was determined. Their theory was that there were two types of metabolism: *anabolism*, the storing up of energy, and *katabolism*, the utilization of stored energy. The determination of sexual characteristics depended on which mode of metabolism prevailed. "In the determination of sex, influences favoring katabolism tend to result in the production of males, as those favoring anabolism

similarly increase the production of females" (Geddes and Thomson 1890, 45, 267). This conclusion was confirmed by looking at the katabolic behavior of adult males (shorter life span, activity and size) compared to the energy-conserving habit of females who they described as "larger, more passive, vegetative, and conservative."¹ In a later review (1914, 205–206) they would say, "We may speak of women's constitution and temper as more conservative, of man's more unstable. . . . We regard woman as being more anabolic, man as relatively katabolic; and whether this biological hypothesis is a good one or not, it certainly does no social harm."

This microcosm/macrocosm relationship between female animals and their nutritive, passive eggs and between male animals and their motile, vigorous sperm was not accidental. Geddes and Thomson viewed the sperm and egg as representing two divergent forms of metabolism established by protozoan organisms, and "what was decided among the prehistoric protozoa cannot be nullified by Act of Parliament." Furthermore, as Aristotle, the difference between the two is nutrition. The motivating force impelling the sperm towards the egg was hunger. The yolk-laden egg was seen as being pursued by hungry sperm seeking their nourishment. The Aristotelian notion of activity and passivity is again linked with the role of the female as nutrient provider. It is also linked with that most masculine of British rituals, the hunt.²

It is usually assumed that the discovery of the X and Y sex chromosomes put an end to these en-

¹The apparent exception of mammalian males was considered due to the extra burden they had when their mates were pregnant.

²Once given "objectivity" by science, the notion that males are active because of their spermatogenic metabolism and women are passive because of their ovum-like ways finds its way in the popular definition of masculinity and femininity. Freud (1933, 175) felt it necessary to counter this view when he lectured on "Femininity": "The male sex-cell is actively mobile and searches out the female one, and the latter, the ovum, is immobile and waits passively. . . . The male pursues the female for the purpose of sexual union, seizes hold of her and penetrates into her. But by this you have precisely introduced the characteristic of masculinity to the factor aggressiveness as far as psychology is concerned." Freud recognized that "it is inadequate to make masculine behavior coincide with activity and feminine with passivity," and that "it serves no useful purpose and adds nothing to your knowledge."

ronmental theories of sex determination. This is today's interpretation and not that of their discoverer. What the genetics texts do not tell us is that C.E. McClung placed his observations of sex chromosomes directly in the context of Geddes and Thomson's environmental model. Using a courtship analogy where in the many spermatid suitors courted the egg in its ovarian parlour, McClung (1901, 224) stated that the egg "is able to attract that form of spermatozoon which will produce an individual of the sex most desirable to the welfare of the species." He then goes on to provide an explicit gender-laden correlation of the germ cells mirroring the behavior of the sexual animals that produced them:

The ovum determines which sort of sperm shall be allowed entrance into the egg substance. In this we see the extension, to its ultimate limit, of the well-known role of selection on the part of the female organism. The ovum is thus placed in a delicate adjustment with regard to the surrounding conditions and reacts in a way to best subserve the interests of the species. To it come two forms of spermatozoa from which selection is made in response to environmental necessities. Adverse conditions demand a preponderance of males, unusually favorable conditions induce an excess of females, while normal environments apportion an approximately equal representation of the sexes. (McClung 1902,76)

McClung concluded this paper by quoting that Geddes and Thomson's theory of anabolism and katabolism provided the best explanation as to whether the germ cells would eventually grow into "passive yolk-laden ova or into minute mobile spermatozoa."

THE SPERM SAGA

Courtship is only one of the narrative structures used to describe fertilization. Indeed, "sperm tales" make a fascinating subgenre of science fiction. One of the major classes of sperm stories portrays the sperm as a heroic victor. In these narratives, the egg doesn't choose a suitor. Rather, the egg is the passive prize awarded to the victor. This epic of the heroic sperm struggling against the hostile uterus is the account of fertilization usually seen in contemporary introductory biology texts.

The following is from one of this decade's best introductory textbooks.

Immediately, the question of the fertile life of the sperm in the reproductive tract becomes apparent. We have said that one ejaculation releases about 100 million sperm into the vagina. Conditions in the vagina are very inhospitable to sperm, and vast numbers are killed before they have a chance to pass into the cervix. Millions of others die or become infertile in the uterus or oviducts, and millions more go up the wrong oviduct or never find their way into an oviduct at all. The journey to the upper portion of the oviducts is an extremely long and hazardous one for objects so tiny. . . . Only one of the millions of sperm cells released into the vagina actually penetrates the egg cell and fertilizes it. As soon as that one cell has fertilized the egg, the (egg) cell membrane becomes impenetrable to other sperm cells, which soon die. (Keeton 1976,394)

We might end the saga by announcing, "I alone am saved." These sperm stories are variants of the heroic quest myths such as the Odyssey or the Aeneid. Like Aeneas, the spermatid hero survives challenges in his journey to a new land, defeats his rivals, marries the princess and starts a new society. The sperm tale is a myth of our origin. The founder of our body is the noble survivor of an immense struggle who deserved the egg as his reward. It is a thrilling and self-congratulatory story.

The details of these fertilization narratives fit perfectly into Campbell's archetype of such myths. Campbell (1956,387), however, believes that "there is no hiding place for the gods from the searching telescope or microscope." In this he has been wrong. The myth lies embedded within microscopic science.³

³There is ample evidence for the ovum as mythic princess. The ovum is not allowed to see sperm before it is of age, and when it travels to meet the sperm this "ripe" ovum not only has a "corona" (crown) but "vestments." It is also often said to have "attendant cells." According to Jung (1967, 171, 204), the hero is the symbol *par excellence* of the male libido and of the longing to reunite with the mother. If true, the sperm is an excellent embodiment of the heroic fantasy. But this does not mean we have to follow this myth. Indeed, one could make a heroic tale about the ovum which has to take a "leap" into the unknown, though its chances of survival are less than 1%. Indeed, the human ovum, too, is a survivor of a process which has windowed nearly all of the original 2 million oocytes, and left it the only survivor of its cohort.

The next passage comes from a book to be given expectant mothers. It, too, starts with the heroic sperm model but then ventures off into more disturbing images.

Spermatozoa swim with a quick vibratory motion. . . . In ascending the uterus and Fallopian tube they must swim against the same current that waft the ovum downward. . . . Although a million spermatozoa die in the vagina as a result of the acid secretions there, myriads survive, penetrate the neck of the uterus and swarm up through the uterine cavity and into the Fallopian tube. *There they lie in wait for the ovum.* As soon as the ovum comes near the *army of spermatozoa*, the latter, as if they were *tiny bits of steel drawn by a powerful magnet, fly at the ovum.* One penetrates, but only one. . . . *As soon as the one enters, the door is shut on the other suitors.* Now, as if *electrified*, all the particles of the ovum (now fused with the sperm) exhibit vigorous agitation. (Russell 1977, 24, emphasis added)

In one image we see the fertilization as a kind of martial gang-rape, the members of the masculine army lying in wait for the passive egg. In another image, the egg is a whore, attracting the soldiers like a magnet, the classical seduction image and rationale for rape. The egg obviously wanted it. Yet, once *penetrated*, the egg becomes the virtuous lady, closing its door to the other *suitors*. Only then is the egg, because it has fused with a sperm, rescued from dormancy and becomes active. The fertilizing sperm is a hero who survives while others perish, a soldier, a shard of steel, a successful suitor, and the cause of movement in the egg. The ovum is a passive victim, a whore and finally, a proper lady whose fulfillment is attained.

The accounts in such textbooks must seem pretty convincing to an outsider. The following is from a paper on the history of conception theories, published—by a philosopher—in 1984.

Aristotle's intuitions about the male as trigger which begins an epigenetic process is a foreshadowing of modern biological theory in which the sperm is the active agent that must move and penetrate the ovum. The egg passively awaits the sperm, which only contributes a nucleus, whereas the egg contributes all the cytoplasmic structures (along with its nucleus) to the zygote. In other words, the egg contributes the material and the

form, and the sperm contributes the activating agent and the form. . . . Thus even modern biology recognizes the specialized and differentiated roles of male and female in an account of conception. Aristotle's move in such a direction was indeed farsighted. (Boylan 1984, 110)

ENERGETIC EGGS AND ACTIVE ANLAGEN

Until very recently, textbook accounts have emphasized (even idealized) the passivity of the egg. The notion of the male semen "awakening the slumbering egg" is seen as early as 1795 (Reil 1795, 79), and this idea, according to historian Tim Lenoir (1982, 37) "was to have an illustrious future." Since 1980, however, there has been a new account of sperm-egg interactions. This revisionism has been spurred on by new data (and new interpretations of old data) which has forced a re-examination of the accepted scenario. The egg appears to be less a "silent partner" and more an energetic participant in fertilization. Two of the major investigators forcing this re-evaluation are Gerald and Heide Schatten. Using scanning electron microscopy, they discovered that when the sperm contacts the egg it does not burrow through.⁴ Rather, the egg directs the growth of microvilli—small finger-like projections of the cell surface—to clasp the sperm and slowly draw it into the cell. The mound of microvilli extending to the sperm had been known since 1895 when E.B. Wilson published the first photographs of sea urchin fertilization. But this structure has been largely ignored until the recent studies, and its role is still controversial.

In 1983, the Schattens wrote a review article for laypeople on fertilization. Entitled "The Energetic

⁴The "burrowing" metaphor is also commonly seen in textbooks, and it brings with it seed-and-soil imagery. This plowing trope was, for many ancient cultures, a metaphor of necessary violence. The active/passive dichotomy is remarkably evident in the verb *to fertilize*. The traditional statement is that the "sperm fertilizes the egg." The sperm is active, the egg is passive. This inverts the original meaning of *fertilize* which involves the nourishment of seeds by the soil. The verb no longer connotes nutrition in this context, but activation.

Egg," it consciously sought to change the metaphors by which fertilization is thought about and taught.

In the past years, investigations of the curious cone that Wilson recorded have led to a new view of the roles that sperm and egg play in their dramatic meeting. The classic account, current for centuries, has emphasized the sperm's performance and relegated to the egg the supporting role of Sleeping Beauty—a dormant bride awaiting her mate's magic kiss, which instills the spirit that brings her to life. The egg is central to this drama, to be sure, but it is as passive a character as the Grimm brothers' princess. Now, it is becoming clear that the egg is not merely a large yolk-filled sphere into which the sperm burrows to endow new life. Rather, recent research suggests the almost heretical view that sperm and egg are mutually active partners. (Schatten and Schatten 1983, 29)

Other studies are showing this mutual activity in other ways. In mammals, the female reproductive tract is being seen as more than a passive or even hostile conduit through which sperm are tested before they can reach the egg. Freshly ejaculated mammalian sperm are not normally able to fertilize the eggs in many species. They have to become *capacitated*. This capacitation appears to be mediated through secretions of the female genital tract. Furthermore, upon reaching the egg, mammalian sperm release enzymes which digest some of the extracellular vestments which surround the egg. These released enzymes, however, are not active. They become activated by interacting with another secretion of the female reproductive tract. Thus, neither the egg nor the female reproductive tract is a passive element in fertilization. The sperm and the egg are both active agents and passive substrates. "Ever since the invention of the light microscope, researchers have marveled at the energy and endurance of the sperm in its journey to the egg. Now, with the aid of the electron microscope, we can wonder equally at the speed and enterprise of the egg, as it clasps the sperm and guides its nucleus to the center" (Schatten and Schatten 1983, 34).

As we have seen above, the determination of maleness and femaleness has also been inscribed by concepts of active masculinity and passive female-

ness. (This means that *sex*, not just *gender*, can be socially constructed!) Indeed, until 1986, all modern biological theories of mammalian sex determination have assumed that the female condition is developed passively, while the male condition is actively produced from the otherwise female state (for review, see Gilbert 1985, 643). This has been based largely on Jost's experiments where rabbits developed the female body condition when their gonadal rudiments were removed before they had differentiated into testes or ovaries. But these experiments actually dealt with the generation of secondary sexual characteristics and not the primary sex determination event—the differentiation of the sexually indifferent gonadal primordia into ovaries or testes.

During the past four years, these theories of primary sex differentiation (notably the H-Y antigen model wherein male cells synthesized a factor absent in female cells which caused the gonadal primordia to become testes) have been criticized by several scientists, and a new hypothesis has been proposed by Eva Eicher and Linda Washburn of the Jackson Laboratory. This new model is based on extensive genetic evidence and incorporates data that could not be explained by the previous accounts of sex determination. In their introductory statement, Eicher and Washburn point out the active and passive contexts that have been ascribed to the development of the primary sexual organs. They put forth their hypotheses as a controlled corrective for traditional views.

Some investigators have over-emphasized the hypothesis that the Y chromosome is involved in testis determination by presenting the induction of testicular tissue as an active (gene directed, dominant) event while presenting the induction of ovarian tissue as a passive (automatic) event. Certainly, the induction of ovarian tissue is as much an active, genetically directed developmental process as is the induction of testicular tissue or, for that matter, the induction of any cellular differentiation process. Almost nothing has been written about genes involved in the induction of ovarian tissue from the undifferentiated gonad. The genetics of testis determination is easier to study because human individuals with a Y chromosome and no testicular tissue or with no Y chromosome and testicular tissue, are relatively

easy to identify. Nevertheless, speculation on the kind of gonadal tissue that would develop in an XX individual if ovarian tissue induction fails could provide criteria for identifying affected individuals and thus lead to the discovery of ovarian determination genes. (Eicher and Washburn 1986, 328)

Again, we see that alternative versions of long-held scientific "truths" can be generated. A feminist critique of cellular and molecular biology does not necessarily mean a more intuitivistic approach. Rather, it involves being open to different interpretations of one's data and having the ability to ask questions that would not have occurred within the traditional context. The studies of Eicher and Washburn on sex determination and those of the Schattens on fertilization can be viewed as feminist-influenced critiques of cell and molecular biology. They have controlled for gender biases rather than let the ancient myth run uncontrolled through their interpretations. Yet the techniques used in their analyses are not different than those of other scientists working in their respective fields, and the approaches used in these studies are no "softer" than those used by researchers working within the traditional paradigms.⁵ . . .

NATURE AS TEXT

"Like other sciences, biology today has lost many of its illusions. It is no longer seeking the truth. It is building its own truth."

—Francois Jacob (1976, 16)

Science is a creative human endeavor whereby individuals and groups of individuals collect data

⁵Although Eicher and Washburn have emphasized that both sexes are actively created, at least two reviews on sex determination have recently proposed one or the other sex as being the "default" condition of the species. It should be noted that the views expressed in this essay may or may not be those of the scientists whose work we have reviewed. It is our contention that these research programs are inherently critical of a masculinist assumption with these respective fields. This does not mean that the research was consciously done with this in mind.

about the natural world and try to make sense of them. Each of the basic elements of scientific research—conceptualization, execution and interpretation—involves creativity. In fact, these three elements are the same as most any artistic, literary or musical endeavor. Two aspects of science are especially creative, namely the conceptual designing of an experiment and the interpreting of the results. Usually, the interpretation is put in the context of a narrative which includes the data but is not dependent upon them (Medawar 1963, 377; Figlio 1976, 17; Landau 1984, 262). Since science is a creative endeavor, it should be able to be criticized as such; and Lewis Thomas (1984, 155) has even suggested that schools of science criticism should exist parallel to that of literary, music and art criticism.

As a creative part of our social structure, biology should be amenable to analysis by feminist critique which has provided new insights into literature, art and the social sciences. Indeed, feminist examinations of sociobiology (Sayers 1982; Bleier 1984) primate research (Haraway 1986), and scientific methods (Keller 1985) have provided an important contribution to the literature of those fields. Researchers in those fields are aware of the feminist criticism and the result has created a better science—one in which methods of data collection and interpretation have been scrutinized for sexual biases.

Any creative enterprise undertaken by human beings is subject to the influences of society. It is not surprising, then, to see how gender becomes affixed to cells, nuclei and even chemicals. Even the interpretations of mathematical equations change with time! The interpretation that Newton gave to his Law of Gravity (i.e., that it was evidence of God's power and benevolence) differs (Dobbs 1985) from the interpretation of eighteenth century physicists (that it was evidence for a mechanical universe devoid of purpose), and from that of contemporary physicists (that it is the consequence of gravitons traversing the curvature of space around matter).

By using feminist critique to analyze some of the history of biological thought, we are able to recognize areas where gender bias has informed how we

think as biologists. In controlling for this bias, we can make biology a better discipline. Moreover, it is important that biology be kept strong and as free from gender bias as possible; for it is in a unique position to do harm or good. As Heschel has remarked (albeit with masculine pronouns):

The truth of a theory about man is either creative or irrelevant, but never merely descriptive. A theory about the stars never becomes a part of the being of the stars. A theory about man enters his consciousness, determines his self-understanding, and modifies his very existence. The image of a man affects the nature of man... We become what we think of ourselves. (1965, 7)

A theory about life affects life. We become what biology tells us is the truth about life. Therefore, feminist critique of biology is not only good for biology but for our society as well. Biology needs it both for itself and for fulfilling its social responsibilities.

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Study and Discussion Questions

1. The Biology and Gender Study Group claims that the active male/passive female model in biology is shaped after social ideals and cultural stereotypes. Examine these ideals or stereotypes by considering particular social activities or roles. That is, what expectations do we have concerning male and female occupations? What is a "women's movie"? a "guy's night out"? girl toys versus boy toys? What messages about gender do advertisements and TV shows send?
2. Are there fields besides cellular biology that assume the active male/passive female model (or soldier/whore, or hero/victim, or prince/sleeping beauty, or any of the other plots or genres described by the Study Group)?
3. In a portion of the essay not included, the Study Group recognizes that the implications of viewing the sperm and egg as married depends on one's view of the marriage relation. They describe marriage-as-domination (modeling the cell after an autocratic Prussian family) and marriage-as-partnership (such as British socialist C. H. Waddington, who married a successful architect and "respected women as intellectual equals"). Describe a range of alternative ways to understand marriage and family relationships. Are these models also applicable to scientific fields besides cellular biology?
4. What aspects of scientific research involve creativity ("are the same as most any artistic, literary, or musical endeavor"), according to the Study Group? Relate these aspects to the standard distinction between context of discovery and justification. Relate them also to Osborne's or Hesse's arguments in chapter 5 on narrative and metaphor.